An improved Grassmannian packing for the cell (4, 64) hlc

Technical note accompanying a Game of Sloanes submission

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1. Summary

This note accompanies the submission of an improved Grassmannian packing for the cell (4, 64) hlc of the Game of Sloanes reference repository. The submitted packing achieves a coherence of μ = 0.687033931214091 (round-8: 0.68703393), strictly below the previously catalogued record of 0.687160201509307 by an absolute gap of approximately -1.263×10^{-4} (−0.0184 %). The packing is the latest in a cascade of four sub-Cohn configurations obtained by the author during May 2026; the cascade is documented in the public repository referenced below.

2. The packing

Record	Date (CEST)	μ (coherence)	Worst pair	Engine
1	12 May 2026 (afternoon)	0.687040494471422	(31, 38)	Aquiles
2	12 May 2026 (evening)	0.687035170223597	(31, 38)	Aquiles
3	16 May 2026 (afternoon)	0.687033937262633	(6, 17)	Boagrius
4	16 May 2026 (evening)	0.687033931214091	(9, 25)	Boagrius

The packing submitted in this PR is Record 4. It consists of 64 unit-norm vectors in $\mathbb{C}^4$, serialised in the standard Game of Sloanes file format: 2·d·n = 512 floating-point values, one per line, real parts first (in row-major order over the 64 vectors of dimension 4), then imaginary parts in the same order. The maximum deviation of any vector from unit norm is 2.22×10^{-16} (machine epsilon). The worst pair is (9, 25), with $|\langle v_9, v_{25}\rangle| = \mu$.

3. Methodology

Records 1 and 2 were produced on 12 May 2026 by an optimisation engine internally named Aquiles, which applies a Riemannian gradient flow with momentum to a one-parameter family of smooth surrogates of the coherence (the water paradigm). Records 3 and 4 were produced on 16 May 2026 by a complementary engine internally named Boagrius, which reformulates the problem as a constrained-feasibility iteration: the coherence is treated as a hard inequality constraint that is progressively tightened, with the engine iterating on the feasibility residual until convergence within a quad-precision tolerance.

The two engines are conceptually complementary. Aquiles is well-suited to finding a new basin from a structurally distinct warmstart; Boagrius is well-suited to deepening a known basin towards the floor of floating-point precision. Records 3 and 4 were obtained by cascading: Boagrius was launched with Record 2 as warmstart, transitioned to a structurally distinct basin during the constraint tightening (worst pair changing from (31, 38) to (6, 17)), produced Record 3, and a second Boagrius run with Record 3 as warmstart converged to Record 4 in a further distinct basin (worst pair (9, 25)). Each Boagrius run took approximately nine minutes of wall time on a single thread of an Apple M2 chip throttled to 25 % CPU.

4. Verification

The packing was cross-validated by five independent code paths, written from scratch with different libraries (and one path with no library at all). The five paths share no implementation between them. All five report identical coherence to the precision available, and all five identify the same worst-pair indices.

#	Kernel	Library / path	μ reported	Worst pair
1	Gram matrix construction	NumPy BLAS matmul	0.6870339312140905	(9, 25)
2	Manual inner product	pure Python, no library	0.6870339312140905	(9, 25)
3	Per-pair inner product	numpy.vdot / LAPACK	0.6870339312140905	(9, 25)
4	Arbitrary precision	mpmath, 50 dps	0.6870339312140905	(9, 25)
5	Arbitrary precision	Python decimal, 40 dps	0.6870339312140905	(9, 25)

The spread across the five kernels is 0.00×10^{0} (byte-exact agreement at 16-digit precision). The worst pair (9, 25) is preserved across all five code paths. The two arbitrary-precision kernels (mpmath at 50 decimal digits, Python decimal at 40 digits of precision) additionally confirm agreement at digit 17 and beyond, well past the precision of float64 arithmetic.

For Record 2 (the predecessor in the cascade), the same five-kernel protocol was applied on 12 May 2026; in addition, the packing was independently reproduced byte-exact by Prof. Henry Cohn (MIT) in his own code 43 minutes after notification, matching all 16 digits and the worst-pair indices (31, 38). The verification discipline used for Record 4 is therefore directly inherited from a protocol that has already been externally validated.

5. Repository and reproducibility

The packing file `4x64_record4.txt`, the other three packings of the cascade, the independent Python verifier (clean-room kernel, no project-internal code paths), the Mac runtime logs, and two technical papers documenting the full methodology are available in a public repository under MIT licence:

`https://github.com/tretoef-estrella/sloane-coherence-records`

A reviewer can verify the submitted packing in under one minute on any standard machine with Python 3 and NumPy:

```
git clone https://github.com/tretoef-estrella/sloane-coherence-records.git
cd sloane-coherence-records
python3 verify_sloane_independent.py 4x64_record4.txt
```

The expected output is `Coherence mu(Phi) = 0.687033931214091`, with worst pair `(9, 25)` and unit-norm check passing within 10^{-10}.

6. Acknowledgments

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Submission of Record 4 ($\mu = 0.687033931214091$) to the Game of Sloanes repository. Companion submissions for the cells (4, 48) hlc and (3, 14) dgm follow in separate pull requests.